

1. KVL states that the algebraic sum of all voltages around a closed loop in a circuit is zero.

It reflects conservation of energy in electrical circuits.

Used in network analysis to solve for unknown voltages and currents.

2.

A transformer is an electrical device that transfers electrical energy between two or more circuits through electromagnetic induction.

Based on Faraday's Law of electromagnetic Induction.

Alternating current in the primary winding creates a varying magnetic flux.

This flux links with the secondary winding ~~creates a varying magnetic flux~~. inducing an electromotive force (EMF).

Works only with AC supply.
Transfers energy without changing frequency, but can step voltage up or down.

3.	Aspect	AC supply	DC Supply
		↓	↓
	Waveform	Alternating sinusoidal	Constant Straight line
	Direction	Changes periodically	Flows in one direction
	Transmission	Efficient for long distance.	Inefficient for long distances
	Generation	Produced by alternators	Produced batteries rectifiers
	Usage	Household power, industries	Electronic portable devices

4. Materials whose electrical conductivity lies between conductors and insulators.

Silicon (Si) Germanium (Ge)

Conductivity increases with temperature.

Can be doped to form p-type or n-type materials.

Basis of modern electronics
(diodes, transistors, ICs).

5. Based on Lorentz force and Fleming's Left-Hand Rule.

Working
Current flows through the armature placed in a magnetic field.

Interaction between magnetic field and current-carrying conductor produces torque.

This torque causes continuous rotation of the motor shaft.

Used in electric vehicles, robotics, fans, and pumps.